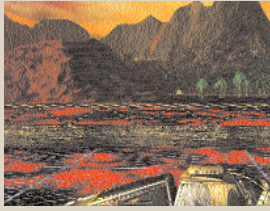


The Games to Look Out for!

Everglade Rush: This futuristic hovercraft racing simu-



lation taps the power of the GeForce3 and coupled with the new features in DirectX 8.0, it features very impressive textures and lighting effects as seen in its pre-release images.

Dronez: This is an RPG that is set in a 3D world and it features some very impres-



sive and breathtaking effects in terms of the complexity and the animations of the 3D models used.

Independence War 2: This space war takes detail to an altogether higher level



where the models boast of polygon counts that near a hundred thousand! These, along with some very impressive lighting and particle effects such as gunfire and explosions, are brought to life with the GeForce3 and DirectX 8.

DIRECTX 8.0

Firstly, what is DirectX? DirectX is an Applications Programming Interface or API. This software is incorporated into Microsoft's operating system and it provides a bridge over which applications and games can access the various features that are supported by multimedia hardware such as 3D accelerators, soundcards, input devices such as joysticks, etc. If a particular 3D graphics card can support a plethora of radically new features, it would amount to nothing if the API does not support these functions.

The primary aim of DirectX was to create a set of tools and interfaces that provides programmers with the building blocks necessary for developing applications that run independent of the hardware upon which these programs are run. This allows developers to concentrate more effort on the features of the application without bothering about hardware issues.

Prior to DirectX, software creators had to develop products to work well on a variety of hardware configurations and there was no real standard for doing this. DirectX provides what is called a Hardware Abstraction Layer (also known as the HAL), which makes use of software drivers to communicate with the application and the hardware.

So if software developers write a code to work with DirectX, they don't need to worry about which hardware platforms it can run on, as all hardware that supports DirectX will run the software.

DirectX provides control to a set of tools that control low-level functions such as 2D graphics acceleration, sound generation mixing and reproduction, and support for input devices such as the mouse and joystick.

All these low-level functions are supported by various components that together make up the foundation of DirectX: Microsoft DirectDraw, Microsoft Direct3D, Microsoft DirectInput, Microsoft DirectSound, Microsoft DirectPlay and Microsoft DirectMusic.

So what's new in DirectX 8.0?

At first glance, you'll find that the new features supported here are strikingly similar to those that have been implemented in the GeForce3. In fact, nVidia has licensed many of its technologies to Microsoft for the development of DirectX 8.0. Little wonder that as of now, GeForce3 is the only graphics processor that is fully DirectX 8.0 compliant. Let's take a look at what new features have been added to DirectX 8.0.

Programmable Vertex Shaders: This feature will allow programmable control over the transform and lighting portion of the graphics processor. Using this, game developers will be

able to simulate far more realistic effects when dealing with objects in skin and cloth and the deformations that are associated with it. Using this feature, developers will be able to impart a different 'feel' to each of their games by making the 3D objects in each behave and react differently.

Programmable Pixel Shaders: Tying in with the feature that is implemented in the Pixel Shader used in cards such as the GeForce3, game developers will now be able to create combined effects such as bump maps with reflections and textures. This would make effects such as water ripples and corrugated metallic surfaces more realistic than ever.

Multi-sample Rendering: In this feature, multiple instances of a 3D scene can be rendered in different frame buffers and are then displayed in the manner desired by the application creator. This would allow for effects such as anti-aliasing, motion blur, depth of field, soft shadows, etc.

Volumetric Textures: This feature ties in with the Programmable Pixel Shader feature. Here, textures will be assigned one more dimension, giving that extra sense of realism.

N-Patches: This allows the complexity of a model to be increased to greater levels. It does this by using special curves called Bezier curves to increase the smoothness in the curved areas of the 3D model. N-Patches could allow a relatively simple model to appear much more detailed.

Texture compression: Textures are one of the biggest elements used in 3D scenes, and they are often the cause of the bottlenecks associated with the graphics card's memory. DirectX 8.0 has incorporated support for texture compression. Compressing all those large texture files will increase the efficiency of the memory bandwidth, resulting in very significant increases in performance and greater visual quality.

Matrix Palette Skinning: Much like vertex skinning, MPS is a feature that allows far more realism in transforming objects that are skinned with textures. Used to generate effects such as wrinkles in cloth or skin, this method provides greater fluidity than vertex skinning as there are more control points used here.

The verdict?

The GeForce3 coupled with DirectX 8.0 promises games that are going to greatly blur the dividing line between games and movies. It looks like the catchword for the next generation of graphics accelerators is not just pure frame rates, but also realism. So the next time you gun the 500 bhp engine in that Lamborghini racing simulation, the visual quality of the game and its special effects might be so intense that you might actually smell the burning rubber! 